Motivations

The development of the Dinacharya Automation system is driven by a convergence of unmet needs in digital wellness, limitations in current AI health tools, and an opportunity to bridge millennia-old Ayurvedic wisdom with modern machine learning. The following motivations collectively justify why this project is both timely and impactful.

M1. The Adherence Crisis in Preventive Health

One of the most well-documented problems in public health is that people consistently fail to maintain beneficial daily habits, even when they understand their importance. This is not a matter of willpower but of design: generic recommendations are not compelling enough to override the friction of daily life.

Consider a simple real-world example: a standard wellness app might remind a user to "exercise for 30 minutes every morning." However, if that user is a Kapha-dominant individual who wakes up sluggish and has back-to-back meetings until 11 AM, that recommendation is both physiologically misaligned and practically impossible. The result is a dismissed notification and a slowly accumulating sense of failure.

Dinacharya, by contrast, is not a one-size-fits-all prescription. It is constitution-aware, time-sensitive, and contextual. The motivation here is to take that rich tradition and make it actionable through automation, closing the gap between knowing what is good for you and actually doing it consistently.

M2. The Digital Health Personalization Gap

The global wellness app market is saturated, yet personalization remains superficial. Most apps adapt to user behavior only in the most basic sense, such as adjusting a step goal upward if you hit it three days in a row. Very few systems attempt deep physiological personalization that accounts for constitutional type, real-time health signals, and cultural health frameworks.

Ayurveda offers a sophisticated model of personalization through the Prakriti-Vikriti framework: your baseline constitution (Prakriti) versus your current state of imbalance (Vikriti).

A Pitta-dominant person in a high-stress week (elevated Vikriti) needs fundamentally different recommendations than the same person on a relaxed weekend. No mainstream wellness app models this today.

For example, on a day when wearable data shows high heart rate variability (HRV) irregularity and broken sleep, the system should recognize this as a Vata-aggravated state and automatically soften the morning routine: replacing vigorous exercise with gentle yoga, swapping cold water for warm herbal tea, and pushing the wake-up nudge 15 minutes later. This kind of physiologically grounded, constitution-aware adaptation is the gap this project directly addresses.

M3. Ayurvedic Knowledge is Rich but Inaccessible in Actionable Form

Classical Ayurvedic texts like the Ashtanga Hridayam and Charaka Samhita contain extraordinarily detailed guidance on daily routines, including specific practices for different times of day, seasons, constitutions, and health states. However, this knowledge exists in Sanskrit texts and translated academic volumes that are not directly usable by a software system or accessible to the average person.

The motivation here is bridging: transforming a text like "Vata individuals should apply warm sesame oil to the scalp and soles before sunrise during Hemanta Ritu (winter)" into a computable rule that fires when the user's Prakriti classification is Vata-dominant, the season is winter, and the current time is pre-sunrise. This kind of operationalization, powered by RAG over a curated Ayurvedic knowledge base, is what makes the system fundamentally different from a generic chatbot that happens to mention Ayurveda.

M4. LLM and Agentic AI Have Just Reached Sufficient Maturity

Until recently, building a system like this was impractical. The components required, including a model that can reason over retrieved text, plan a daily schedule, adjust based on physiological signals, generate natural language explanations, and handle safety overrides, did not exist in a usable, integrated form.

The emergence of LangGraph for multi-agent orchestration, FAISS/Weaviate for semantic vector retrieval, and instruction-tuned LLMs capable of structured planning means this is now feasible as a capstone project. The system can be prototyped with real components rather than simulated behavior. This is a narrow window of technological opportunity: the tools exist, the research foundations are being laid, and no production system has yet combined all these elements in a dinacharya-specific context.

As a concrete example: the Plan-and-Execute agent pattern used in this project was not practically available before LangGraph matured. Earlier ReAct-style agents were too brittle for the kind of multi-step daily schedule generation the system requires. The timing of this project aligns directly with when the technology crossed the feasibility threshold.

M5. Safety and Hallucination Risks Demand a Grounded Architecture

A powerful but underappreciated motivation is the risk side of the problem. LLMs, when applied to health-adjacent domains without grounding, can hallucinate authoritative-sounding but incorrect or even harmful recommendations. A naive implementation might suggest Abhyanga (oil massage) for a user who has just reported a fever, or recommend vigorous Vyayama (exercise) immediately after a meal, both of which are explicitly contraindicated in classical Ayurvedic literature.

The motivation to build a retrieval-augmented, rule-grounded, safety-checked system is therefore not just about accuracy but about responsibility. By encoding clinical contraindications as hard symbolic overrides that no LLM output can bypass, the system establishes a safety contract. This architecture is a direct response to documented failure modes in health AI and is part of what makes this project publishable and ethically sound.

M6. Preventive Health Has Massive Scale Potential

Non-communicable diseases (NCDs) such as type 2 diabetes, hypertension, sleep disorders, and metabolic syndrome are heavily influenced by daily behavioral patterns. These are conditions where consistent lifestyle change, if maintained, can rival pharmaceutical intervention in effectiveness. Yet the healthcare system has almost no infrastructure for sustained daily behavioral support outside of occasional clinical visits.

An autonomous dinacharya agent running on a smartphone, continuously adapting to the user's physiological state and nudging them toward health-promoting behaviors, represents a scalable public health intervention. The system requires no clinic visit, no prescription, and minimal user effort once initialized. If the pilot (n=20) shows even modest adherence improvement, the underlying model has enormous potential for larger deployment, especially in populations where Ayurvedic medicine is culturally resonant.

Objectives

The following objectives define the concrete, measurable goals of the Dinacharya Automation project. They are organized to reflect the project's dual nature as both a technical engineering effort and a preliminary clinical feasibility study.

O1. Design and Implement a Dosha-aware Personalization Pipeline

The first and foundational objective is to construct a machine learning pipeline that can reliably classify a user's Ayurvedic constitution and detect real-time deviations from it.

- **Prakriti Classification:** Train a Random Forest or XGBoost ensemble on a 24-item bilingual phenotypic questionnaire covering body frame, skin type, digestion, sleep, and temperament. The model must output a normalized tri-dosha vector [V, P, K] with confidence scores. Target classification agreement of >=75% with expert Ayurvedic practitioner assessments on a held-out validation set.
- **Vikriti Anomaly Detection:** Implement an unsupervised Isolation Forest model on rolling wearable data (7-day window of HRV, sleep stages, resting HR) that fires an alert when a user's current state deviates significantly from their Prakriti baseline. For example, a Pitta-baseline user showing Vata-like HRV irregularity and disrupted sleep should trigger a Vata-aggravation flag that shifts the day's recommendations accordingly.
- **Multimodal Feature Engineering:** Build preprocessing pipelines that transform raw wearable signals (RMSSD, SDNN for HRV; sleep stage durations; step counts) and contextual data (GPS-derived season/weather, calendar stress events) into Ayurvedic proxy features usable by downstream models.

O2. Build a Retrieval-Augmented Generation (RAG) Knowledge Base

To ensure recommendations are grounded in authentic Ayurvedic tradition and not hallucinated, the system requires a curated, queryable knowledge base.

- **Knowledge Curation:** Collect, clean, and chunk Ayurvedic guidelines from vetted sources into discrete, semantically coherent units (e.g., a single chunk might be: 'Vata individuals should avoid dry, cold, and light foods during Vata season; prefer warm, oily, and nourishing meals'). Each chunk is tagged with metadata: dosha relevance, season, time-of-day, activity type, and contraindication flags.
- **Vectorization and Indexing:** Embed all chunks using a sentence transformer model and index them in FAISS or Weaviate for efficient approximate nearest-neighbor retrieval. The target is sub-200ms retrieval latency for real-time planning.
- **Retrieval Accuracy:** Evaluate retrieval quality on a test set of 50 clinical queries (e.g., 'What morning practices are recommended for a Vata-Pitta individual in summer with poor sleep?'). Target a Precision@5 of >= 0.80, meaning at least 4 of the top 5 retrieved chunks should be assessed as relevant by an Ayurvedic expert reviewer.

O3. Develop the LLM Planning Agent with Safety Guardrails

The cognitive core of the system is an LLM-based Plan-and-Execute agent that synthesizes the user's dosha state, retrieved knowledge, and contextual constraints into a usable 24-hour schedule.

- **Planner Agent:** Implement a LangGraph-orchestrated Planner Agent that receives the tri-dosha vector, current Vikriti flags, retrieved Ayurvedic guidelines, user calendar data, and weather context. The agent must decompose the day into Morning (Brahma Muhurta to mid-morning), Midday, and Evening blocks, assigning specific practices to each block with time estimates and brief natural language rationale.
- **Executor Agent:** Implement an Executor Agent that calls external tools (weather API, calendar integration) to fine-tune timing. For example: if the calendar shows a 7 AM meeting, the executor reschedules the Nasya practice from 6:30 AM to post-meeting, checking against contraindications before confirming.
- **Safety Engine (Hard Overrides):** Implement a symbolic rule engine that runs as a mandatory final filter on every generated schedule. The engine enforces at minimum

the following contraindications from classical Ashtanga Hridayam: (a) Abhyanga is blocked during acute fever, indigestion, or menstruation; (b) Vyayama is blocked immediately post-meal or during emaciation; (c) Nasya is blocked post-meal; (d) Dantadhavana is blocked during vomiting or throat conditions. Any schedule element flagged by the safety engine is automatically removed and replaced with a safe alternative.

- **Natural Language Explanation:** Every recommendation must be accompanied by a user-facing rationale in plain language (e.g., 'You have a Vata-aggravated morning based on last night's irregular sleep. We've swapped your jog for a 20-minute walk and added warm sesame oil application to help ground and calm your nervous system.'). This explainability is critical for trust and adherence.

O4. Implement Closed-Loop Behavioral Adaptation

A static schedule is insufficient; the system must learn from the user's actual behavior over time and adapt its recommendations accordingly.

- **Adherence Tracking:** Implement an Adherence Score (S_adj) computed as the ratio of completed tasks to recommended tasks over a rolling 7-day window, updated daily. Users mark tasks complete via the mobile app; passive inference from wearable data (e.g., detected walking pattern implying morning yoga was done) supplements self-report.
- **Adaptive Complexity Scaling:** When S_adj falls below 0.5, the agent automatically simplifies the routine to 3-4 'Anchor Habits' (typically meal timing, hydration, and a single morning practice). When S_adj exceeds 0.8, the agent introduces advanced Sattva-enhancing practices such as Trataka (candle gazing), specific Pranayama sequences, or Marma point self-massage.
- **Nudging System (EAST Framework):** Deploy behavior change nudges aligned with the EAST principles: Easy (pre-scheduling activities as calendar defaults), Salient (framing notifications around immediate benefits like 'Improve your digestion today' rather than abstract Ayurvedic terminology), and Timely (triggering nudges during peak smartphone interaction windows identified from usage patterns).

O5. Pilot Study Evaluation (n=20)

Beyond building the system, the project includes a structured empirical evaluation to assess preliminary efficacy. This distinguishes the work as a research contribution rather than just an engineering prototype.

- **Participants and Design:** Recruit 20 participants for a 4-week pilot study. Participants complete an initial Prakriti assessment, receive the personalized dinacharya system on their smartphone with wearable integration, and are followed for 4 weeks with weekly check-ins. A control condition (generic wellness app) will be compared where feasible.

- **Primary Outcome - Adherence:** Measure weekly S_adj scores. Hypothesis: mean adherence will be >= 0.65 by week 4, compared to < 0.30 observed in comparable generic wellness app studies. This would validate that dosha-personalized, contextually adaptive scheduling meaningfully outperforms generic reminder systems.

- **Secondary Outcome - Physiological Proxies:** Measure changes in objective wearable metrics from baseline to week 4: sleep duration (target: >= 15 minute average increase), RMSSD-based HRV (target: measurable improvement in Vata and Pitta dominant users), and resting heart rate. These are not primary efficacy outcomes (the pilot is underpowered for this) but serve as preliminary signals for future clinical study design.

- **Usability Evaluation:** Administer the System Usability Scale (SUS) at weeks 2 and 4. Target a SUS score >= 70 (classified as 'Good' usability), indicating the system is ready for larger deployment. Qualitative interviews will probe the perceived relevance of recommendations, trust in the system, and barriers to adherence.

- **Ablation Study:** Run a computational ablation comparing three system variants: (1) Full system with RAG + dosha personalization + safety engine; (2) LLM planning without RAG (no knowledge grounding); (3) Fixed rule-based schedule without LLM. Compare schedule quality ratings from an Ayurvedic expert reviewer. This quantifies the contribution of each architectural component.

O6. Document Safety, Privacy, and Clinical Translation Pathways

Given that this system operates in a health-adjacent domain and processes sensitive biometric and behavioral data, a responsible engineering objective is to explicitly address safety, privacy, and the roadmap for any future clinical deployment.

- **Privacy Architecture:** All user biometric data is stored locally on-device or in an encrypted PostgreSQL instance with no third-party data sharing. The Prakriti and Vikriti vectors are pseudonymized. The system complies with the principle of data minimization, collecting only signals required for the dosha inference and scheduling pipeline.
- **Safety Documentation:** Publish a Contraindication Registry documenting every safety rule in the clinical override engine, with citations to classical Ayurvedic sources and any available modern clinical literature supporting the contraindication. This registry must be reviewed by at least one qualified Ayurvedic practitioner before the pilot deployment.
- **Scope Boundaries:** Explicitly document what the system does not do: it does not diagnose disease, recommend pharmaceutical interventions, serve acute or emergency health needs, or replace clinical Ayurvedic consultation. All user-facing communications include a clear disclaimer. This scope documentation is a prerequisite for ethical IRB approval of the pilot study.

Summary

Together, these six objectives define a coherent research and engineering program that advances both the science of personalized wellness AI and the practical availability of Ayurvedic daily routine guidance. Success on all objectives would constitute a publishable contribution demonstrating the feasibility of dosha-aware, agentic automation of traditional health practices in a modern digital health context.

Lit survey

LITERATURE SURVEY

Section 1 — Prakriti computational modeling

An automated system for Ayurvedic Prakriti classification using machine learning and deep learning techniques" (Singh et al., 2023)

(https://link.springer.com/article/10.1007/s00500-023-07942-2)

- **Objective:** To develop an automated system for classifying human Prakriti (Vata, Pitta, Kapha) by comparing traditional Machine Learning (ML) with modern Deep Learning (DL) architectures.
- **Key Findings:** The study demonstrates that **Deep Learning (specifically Multi-layer Perceptrons)** significantly outperforms standard ML models in classification accuracy for Ayurvedic constitutions.
- **Significance:** It highlights the transition from simple decision-tree-based logic to more complex neural networks for handling Ayurvedic data.

  Limitations:

  - **Static Input:** The model relies on **static questionnaire data**, which captures a "snapshot" in time rather than the user's evolving physiological state.
  - **Classification Only:** It focuses solely on **categorizing** the user into a Prakriti type but does not provide actionable, time-sensitive recommendations or scheduling.
  - **Lack of Context:** It does not account for **Vikriti** (current imbalances caused by environmental factors, stress, or poor sleep).

Artificial intelligence for personalized nutrition: A review of tools and applications

(https://onlinelibrary.wiley.com/doi/10.1111/coin.70133)

- **Objective:** To review how AI, specifically Machine Learning (ML) and Deep Learning (DL), can process complex data (biological, behavioral, and environmental) to provide tailored nutritional and health advice.
- **Key Findings:** The paper highlights that AI is superior to traditional methods for managing "high-dimensional" data—like combining blood markers with wearable data—to predict individual responses to lifestyle choices.

Limitations of the Paper

- **Narrow Scope:** It focuses primarily on **nutrition**
- **Lack of Traditional Knowledge:** The paper relies on Western biological markers (genomics, microbiome) and lacks **culturally grounded domain knowledge** like the **Prakriti and Vikriti** assessments.
- **Static vs. Agentic:** It reviews systems that mostly provide static advice.

A Survey on Machine Learning Techniques for Human Prakriti Classification
(Gupta et al., 2019) (https://ieeexplore.ieee.org/document/9057416)

- **Objective:** To review and compare various machine learning algorithms (Random Forest, SVM, Naive Bayes, Decision Trees) used by researchers to classify individuals into Vata, Pitta, or Kapha types.
- **Key Finding:** The paper highlights that while multiple ML methods are effective, the accuracy depends heavily on the quality of feature extraction from Ayurvedic questionnaires.

Limitations

- **Non-Empirical:** As a survey, it lacks original experimental data or a new functional prototype.
- **Static Scope:** It focuses entirely on **static classification** via questionnaires and does not address the **real-time multimodal sensing** (HRV, sleep) that is central to your project.
- **No Actionable Output:** It discusses how to *identify* a Dosha but offers no framework for *acting* on that data, such as the **agentic planning and scheduling** of Dinacharya routines

Analysis and Synthesis of A Human Prakriti Identification System Based on Soft Computing Techniques(Gupta et al., 2019/2020)
(https://www.eurekaselect.com/article/98303)

- **Objective:** To eliminate the subjectivity and time-intensity of traditional Ayurvedic Prakriti assessment by using Machine Learning to automate the diagnostic process.
- **Key Finding:** The study proves that machine learning algorithms can successfully identify a user's dominant Dosha (Vata, Pitta, or Kapha) by analyzing specific physiological and psychological features.
- **Data Collection:** Data is gathered using a **standardized Ayurvedic questionnaire** (often based on Ayusoft or similar clinical tools).
- **Features:** The dataset includes **150+ features** covering physical attributes (e.g., skin texture, hair type), physiological habits (e.g., appetite, sleep patterns), and psychological traits (e.g., temperament, memory).

Limitations

- **Small Dataset:** The relatively small sample size may limit the model's ability to generalize across diverse populations.
- **Subjectivity of Self-Reporting:** Relying on questionnaires introduces "self-report bias," which your project aims to mitigate by using **"multimodal sensing"** and objective **"physiological proxies"** like HRV.
- **Static Assessment:** The paper only identifies a user's *Prakriti* (birth constitution) and does not address **"Vikriti"** (current imbalances)

Hybrid Machine Learning for Personalized Ayurvedic Drug Recommendation via Prakriti Assessment (IEEE, 2025)

Objectives: The researchers developed a **hybrid system** that combines clustering, classification, and natural language processing (NLP) to avoid the high computational costs of deep learning while remaining interpretable.

- **Prakriti Classification:** Uses **Random Forest** to classify users into Vata, Pitta, or Kapha types based on physical and psychological traits.
- **Patient Segmentation:** Employs **K-Means Clustering** to group patients with similar profiles for tailored treatment plans.
- **Symptom Processing:** Uses an **NLP Engine** (BERT or TF-IDF) to analyze unstructured text from patient symptoms and feedback.
- **Rule-Based Integration:** Combines these ML outputs with a **traditional Ayurvedic rule-based system** to generate final drug and lifestyle recommendations.

Key Findings

- **High Accuracy:** The Random Forest model achieved **95% accuracy** in classifying Prakriti types.
- **Scalability:** The hybrid approach proved effective at delivering individualized care without the overhead of complex neural networks

Dataset Details

- **Size:** Based on the results, the system was tested on a large dataset of approximately **15,000 samples** for Prakriti classification (roughly 5,000 per Dosha) and **10,000 samples** for the NLP engine.
- **Features:** Data included **demographics** (age, gender, body type), **physical traits** (skin, hair, eyes, digestion), **lifestyle data** (diet, exercise, stress), and **unstructured feedback/symptoms**

Limitations

- **Manual Feature Engineering:** The system relies on structured profile data which still requires significant pre-processing and manual input.
- **Initial Feedback Loop Issues:** The paper's results show that the **feedback loop** for retraining models is in its early stages, with low effectiveness scores (0.50 accuracy) for tracking treatment outcomes.
- **Lack of Real-Time Data:** The researchers explicitly state that a future enhancement would be the inclusion of **real-time monitoring**, as the current recommendations are based on static inputs.

Section 2 — Multimodal health sensing

AyuRAG: Ayurvedic Knowledge Integration with Retrieval-Augmented Lightweight Large Language Models

- **Objective: Objective:** To eliminate the subjectivity and time-intensity of traditional Ayurvedic Prakriti assessment by using Machine Learning to automate the diagnostic process.

- **Key Finding:**
    - RAG improves factual accuracy: Using retrieval-augmented generation allows the LLM to generate responses **grounded in external knowledge**, reducing hallucination.
    - Lightweight LLMs are sufficient: The system demonstrates that **smaller LLMs combined with retrieval systems** can still provide meaningful responses.
- **Data Collection:** The system uses a **curated Ayurvedic knowledge base** rather than a traditional machine learning dataset.

AI-Driven Personalized Healthcare: Leveraging Multimodal Data for Precision Medicine

- The paper proposes an AI-based framework that integrates multiple healthcare data sources such as physiological signals, medical records, and contextual patient data to support personalized healthcare decisions. By combining heterogeneous data sources, AI models can identify patterns that improve diagnosis and treatment personalization.
- **Limitations**
    - No single real-world dataset used for evaluation.
    - The framework is conceptual rather than fully implemented.
    - Does not integrate conversational AI or LLM systems.
    - No adaptive behavioral feedback loop.
- **Key Findings**
    - Multimodal data integration significantly improves personalized healthcare predictions.
    - AI can analyze heterogeneous health data streams more effectively than traditional single-source systems.
    - Combining wearable data, clinical records, and contextual information improves healthcare decision-making.
    - Multimodal AI systems are a key enabler of precision medicine.
- **Data Sources Used**
    - The system integrates multiple healthcare data modalities:
    - wearable sensor data (heart rate, activity, sleep)
    - electronic health records
    - clinical test results

AI-Driven Multimodal Workflow Optimization for Personalized Patient-Centered Care (DOI: https://doi.org/10.30574/wjaets.2025.15.2.0604)

- **Objective:** To explore how artificial intelligence can optimize healthcare workflows by integrating multiple sources of patient data and enabling personalized patient-centered care. The study examines AI-driven systems that combine heterogeneous healthcare data to improve clinical decision-making and treatment planning.
- **Key Findings:**

  • **Multimodal data improves healthcare insights**: Integrating multiple healthcare data sources allows AI systems to generate more accurate predictions and better patient-specific treatment recommendations.

• **AI improves healthcare workflow efficiency:** AI-driven analytics can streamline clinical processes, reduce delays in diagnosis, and support healthcare professionals with decision-support tools.

• **Patient-centered care becomes more feasible**: Combining clinical records, sensor data, and contextual information enables systems to tailor healthcare services to individual patients.

- **Data Collection**
    - The paper discusses integrating heterogeneous healthcare data such as:
    - electronic health records (EHR)
    - physiological sensor data
    - patient medical history
    - contextual healthcare information
- **Limitations**
- Review-style study
- No specific dataset
- No personalized health agent
- Limited technical implementation

Human Activity Recognition with Smartphones using Machine Learning Algorithm (DOI: https://doi.org/10.1109/ICKECS61492.2024.10617109)

Objective: Develop a machine learning framework that recognizes human activities using smartphone sensor data.

Dataset: Smartphone sensor data including accelerometer, gyroscope, and magnetometer signals.

Methods: Feature extraction and machine learning classification for activity recognition.

Key Findings: Smartphone sensor data can accurately detect human activities and support context-aware monitoring systems.

Limitations: No personalized health recommendations or adaptive behavior modeling.

Section 3 — LLM healthcare agents

GatorTronGPT: A Large Language Model for Clinical Natural Language Processing
(https://www.nature.com/articles/s41746-023-00844-6)

Objective: To develop a **domain-specific large language model trained on large-scale clinical data** that can perform medical question answering, clinical reasoning, and knowledge retrieval tasks using real healthcare data sources.

Dataset: The model was trained on **very large clinical datasets**, including:

• **90+ billion tokens** from clinical narratives
• biomedical literature (PubMed)
• medical textbooks
• healthcare records

Key Findings

• **Domain-specific training significantly improves medical reasoning.**
• The model achieves **state-of-the-art results on several clinical NLP benchmarks.**
• Retrieval-based knowledge grounding improves answer reliability.
• LLM systems trained on clinical data can assist healthcare professionals with decision support.

Limitations

• Requires extremely large clinical datasets for training.
• High computational cost for training and inference.
• Not designed specifically for personalized patient monitoring systems.
• Clinical deployment requires strong privacy and safety controls.

Conversational Health Agents: A Personalized LLM-Powered Agent Framework (https://academic.oup.com/jamiaopen/article/8/4/ooaf067/8186991)

Objective: To design a **general framework for building LLM-based conversational health agents (CHAs)** that can provide personalized healthcare assistance by integrating external data sources, knowledge bases, and analysis tools.

Key Findings

• **LLM orchestrator enables agent behavior**
The system uses an orchestrator that analyzes user queries, gathers information from external sources, and generates personalized responses.

• **Integration with external knowledge sources**
The agent framework connects to APIs, healthcare databases, and machine-learning models to improve reliability.

• **Multimodal interaction capability**
The system supports multiple forms of input such as text, audio, and images for healthcare interaction.

Data Collection:

- healthcare knowledge bases
- external APIs

- AI models for health analysis
- conversational user data

Limitations

• Framework-level study rather than a single deployed system
• Limited real-world clinical validation
• Performance depends on external data sources

Arkangel AI: A Conversational Agent for Evidence-Based Medical Question Answering (https://doi.org/10.1016/j.ibmed.2025.100274)

Objective

To build an LLM-based conversational agent capable of answering medical questions using a multi-LLM workflow with evidence retrieval from medical literature.

Key Findings

• **Multi-LLM architecture improves accuracy**
The system uses five LLM components that work together to retrieve evidence and generate answers.

• **Strong performance on medical benchmarks**
Accuracy reached **~90% on MedQA dataset**, outperforming several baseline models.

• **Effective retrieval-augmented reasoning**
The system retrieves medical literature before generating responses, improving factual reliability.

Data Set:

- **MedQA dataset**
- **PubMedQA dataset**

These contain medical question-answer pairs derived from clinical exam material and research literature.

Limitations

• Requires multiple LLM components (high computational cost)
• Final medical decisions still require human experts
• Focused mainly on medical question answering

Healthcare Agent: Eliciting the Power of LLMs for Medical Consultation (https://www.nature.com/articles/s44387-025-00021-x)

Objective: To design a healthcare conversational agent capable of performing **multi-turn medical consultations** using LLM reasoning and structured dialogue management.

Key Findings

• **Dialogue planning improves medical conversations**
The system plans multi-turn conversations with patients to gather relevant clinical information.

• **Memory module improves patient interaction**
Conversation history and medical records are stored to maintain context.

• **Safety module improves reliability**
The agent includes mechanisms to detect unsafe medical responses and enforce ethical constraints.

Data Collection

- simulated patient interactions
- medical consultation datasets
- clinical dialogue examples

Limitations

• Requires physician oversight for final decisions
• Evaluation mostly based on simulated patient interactions
• Limited integration with wearable or physiological data

Personal Health Insights Agent (PHIA): LLM Agents for Wearable Health Data Analysis (https://www.nature.com/articles/s41467-025-67922-y)

Objective: To build an LLM-based agent that can analyze personal wearable health data and generate personalized health insights by performing multi-step reasoning and data analysis on behavioral health signals.

Data Sources: The system uses large-scale wearable datasets, including:

- sleep data
- physical activity
- heart rate signals
- behavioral health metrics

The authors also released:

synthetic wearable dataset
4000+ health question dataset

Key Findings

• **LLM agents can analyze wearable health data effectively**
Agent-based reasoning allows LLMs to process complex time-series wearable data.

• **Personalized health insights are possible**
The system generates individualized health explanations rather than generic responses.

• **Agent architecture improves reasoning ability**
Multi-step reasoning enables more accurate analysis of behavioral health patterns.

• **LLM agents outperform standard LLM models**
Agent-based frameworks perform better for wearable health queries.

Limitations

• Focuses mainly on **wearable behavioral data**, not clinical health records.
• Does not incorporate **real-time physiological monitoring pipelines**.
• Still requires **large computational resources for agent reasoning.**
• System focuses on **analysis and insight generation**, not direct medical diagnosis

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Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks
(https://arxiv.org/abs/2005.11401)

Objective: To improve factual accuracy of language models by combining **neural text generation with external document retrieval**, enabling models to access large knowledge bases during inference instead of relying only on parameters stored during training.

Dataset: The model was evaluated on several **knowledge-intensive NLP datasets**, including:

- Natural Questions
- WebQuestions
- TriviaQA
- FEVER fact-checking dataset

Key Findings

• **External knowledge improves factual accuracy**
Language models perform better when they retrieve relevant documents before generating responses.

• **RAG reduces hallucinations**
Grounding answers in retrieved documents improves reliability.

• **End-to-end retrieval + generation improves QA performance**
The combined retriever-generator architecture outperforms traditional QA systems.

• **Dynamic knowledge updates**
The system can update knowledge simply by updating the document database.

Limitations

• Retrieval quality strongly affects final answers.
• Dense retrieval systems require large vector indexes.
• Latency increases due to retrieval steps.
• Still relies on language model reasoning ability.

MedRAG: Enhancing Retrieval-augmented Generation with Knowledge Graph-Elicited Reasoning for Healthcare Copilot
(https://arxiv.org/abs/2502.04413)

Objective: To improve medical decision-support systems by integrating **retrieval-augmented generation with a diagnostic knowledge graph**, enabling LLMs to reason over clinical symptoms and medical knowledge when generating responses.

Dataset:

• **DDXPlus dataset** (public differential diagnosis dataset)
• **CPDD dataset** (chronic pain dataset collected from Tan Tock Seng Hospital)

Key Findings

• Knowledge-graph reasoning improves diagnostic accuracy.
• Combining EHR retrieval with structured knowledge reduces misdiagnosis.
• The system outperforms existing RAG medical models in diagnostic specificity.
• The agent can generate **follow-up diagnostic questions** to refine predictions.

Limitations

• Requires high-quality medical knowledge graphs.
• Access to EHR datasets may be restricted due to privacy concerns.
• Computational complexity increases with graph reasoning.

System Desgin

1.1 Neuro-Symbolic Hybrid Framework

The system follows a neuro-symbolic design that bridges data-driven machine learning with logic-based Ayurvedic principles.[1] The architecture is divided into three primary functional domains:

- **Perception Domain:** Utilizes neural networks and ensemble models to process noisy, high-frequency wearable data and questionnaire-based phenotypic traits.[2]
- **Cognitive Planning Domain:** Employs a Large Language Model (LLM) agent to reason over retrieved knowledge and user context.
- **Guardrail Domain:** A symbolic rule engine that enforces strict clinical contraindications, ensuring safety and traditional authenticity.[4]

1.2 Logical System Flow

The system operates as a closed-loop pipeline:

1. **Ingestion:** Multimodal data (wearables, context, symptoms) is ingested.
2. **Phenotyping:** The system computes the user's $Prakriti$ (static) and $Vikriti$ (dynamic) vectors.[5]
3. **Retrieval:** Relevant Ayurvedic guidelines are pulled from a vector database (RAG).
4. **Planning:** The Agentic Planner generates a 24-hour schedule.[6]
5. **Execution & Nudging:** The Schedule Generator delivers context-aware nudges.[7]
6. **Adaptation:** The system monitors adherence and modifies future plans.[8]

2.1 Multimodal Ingestion Layers

The system integrates four distinct categories of data to build a comprehensive "Computable Phenotype":

Data Category	Source	Metrics/Features	Ayurvedic Proxy
Biometric	Wearables	HRV (RMSSD, SDNN), Sleep Stages, Resting HR.	Autonomic balance (Dosha flux).[10]
Phenotypic	Questionnaire	24-item bilingual assessment (Body	$Prakriti$ (Constitutional

		frame, skin, digestion).	baseline).[12]
Contextual	Smartphone	GPS (Season/Weather), Calendar (Stress events).	$Ritucharya$ (Seasonal alignment).[13]
Subjective	Self-Report	Cravings, energy levels, digestive discomfort.	$Vikriti$ (Transient imbalance).[5]

2.2 Physiological Signal Processing

- **HRV Analysis:** The system maps heart rate variability to dosha states. Vata-dominant states are characterized by high but irregular variability, Pitta by intermediate variability, and Kapha by stable, periodic rhythms.[10]
- **Sleep Metrics:** Sleep duration and latency are used to predict dosha aggravation. Higher Vata scores significantly predict a longer time to fall asleep ($p < 0.01$) and reduced morning restedness ($p < 0.001$).[14] Kapha-dominant states correlate with daytime somnolence and longer naps.[14]

3.1 Dosha Mapping Models

The system employs dual machine learning pipelines for phenotype classification:

- **Prakriti Classification:** A Random Forest or XGBoost ensemble model trained on phenotypic features. Random Forest is prioritized for its ability to handle high-dimensional, non-linear relationships between Ayurvedic traits.[2]
- **Vikriti Anomaly Detection:** An unsupervised learning module (e.g., Isolation Forest) that identifies deviations from the user's $Prakriti$ baseline using real-time wearable data.[5]

3.2 Knowledge Retrieval-Augmented Generation (RAG)

To ground the LLM in authentic tradition, the system uses a vector database (FAISS or Weaviate) containing indexed classical texts.

- **Vectorization:** Ayurvedic rules are chunked and stored as embeddings (e.g., "Exercise in morning reduces Kapha").
- **Retrieval Logic:** When $Vikriti$ is detected (e.g., $+Vata$), the system queries the

database for Vata-pacifying rituals specific to the current $Ritu$ (season).[13]

4.1 Planning Agent Architecture

The system utilizes a **Plan-and-Execute** agentic pattern rather than a standard ReAct loop.[17]

- **Planner Agent:** Receives the tri-dosha vector $[V, P, K]$, user schedule, and retrieved rules to decompose the day into subtasks (Morning, Midday, Evening).[6]
- **Executor Agent:** Triggers specific tools (e.g., "Check Weather," "Verify Calendar") to fine-tune the timing of each activity.[18]

4.2 Rule-Based Safety Engine (Clinical Overrides)

Every recommendation must pass through a symbolic safety layer before delivery.[2] This layer implements hard-coded clinical contraindications:

Activity	**Contraindications (Safety Override)**	**Rationale**
Abhyanga	Acute Fever (*Nava Jwara*), Indigestion (*Ajirna*), Menstruation.	Prevents lodging toxins (*Ama*) deeper into tissues.[19]
Vyayama	Emaciation (*Kshaya*), Severe Indigestion, immediately after meals.	Prevents $Vata$ aggravation and digestive disruption.[21]
Nasya	Immediately after meals or drinking large amounts of water.	Risk of sinus irritation and $Agni$ (digestive) weakening.[22]
Dantadhavana	Vomiting, cough, indigestion, or diseases of the throat.	Avoids aggravating acute upper respiratory conditions.[25]

5.1 Adherence & Closed-Loop Adaptation

The system monitors user engagement through an **Adherence Score (S_{adj})**, calculated as the ratio of completed to recommended tasks over a rolling 7-day window [26]:

$$S_{adj} = \frac{\sum \text{ActualUse}}{\sum \text{IntendedUse}}$$

- **Low Adherence ($S_{adj} < 0.5$):** The planning agent reduces the routine's complexity, prioritizing only the most critical "Anchor Habits" (e.g., meal timing).[8]
- **High Adherence ($S_{adj} > 0.8$):** The agent introduces advanced "Sattva-enhancing" practices (e.g., specific Pranayama or meditation).[28]

5.2 Nudging Framework (EAST/MINDSPACE)

The schedule delivery uses digital nudges to influence behavior without restricting choice [7]:

- **Easy (E):** Activities are pre-scheduled as "Defaults" based on the Ayurvedic clock (e.g., 6 AM–10 AM for Kapha-clearing exercise).[7]
- **Salience (S):** Notifications highlight immediate benefits (e.g., "Improve digestion" vs. "Reduce Vata").[7]
- **Timely (T):** Nudges are triggered during peak receptivity, identified via smartphone interaction patterns.[7]

5.3 Technical Implementation Stack

- **Backend:** Python/FastAPI with LangGraph for agent orchestration.[32]
- **ML/NLP:** Scikit-learn for $Prakriti$ classification; Hugging Face/Gemini for reasoning.[4]
- **Database:** PostgreSQL for user logs; FAISS/Weaviate for vector storage.[32]
- **Mobile Client:** React Native to support cross-platform wearable integration (Apple HealthKit/Google Fit).[30]

System Design + Architecture: